Department of Computer Science and Engineering
Shiv Nadar University Chennai
July-November 2026 Semester
M.Tech. Course on Deep Learning
Lab Assignment 5

Date: 28th September, 2026

1. **Build a CNN model for image classification. The CNN model should have two convolution layers and two pooling layers. The layers are denoted by CL1, PL1, CL2 and PL2, where CL stands for a convolution layer and PL stands for a pooling layer. In the convolution layers, the filter size F is 3 and the stride factor S is 1. The number of feature maps is 4 in CL1 and 6 in CL2. The ReLU is used as the activation function in the convolution layers. The pooling layers use max pooling with the filter size F of 2 and the stride factor S of 2. Use the softmax as the activation function in the output layer. Mini-batch mode with the number of subsets as 5 is used for training. The AdaM method is the optimization method.**
   (a) **Train the model without batch normalization**
   (b) **Train the model with post-activation batch normalization**
2. **Use the transfer learning method to build a model for image classification. Use the VGGNet as the pre-trained model. Perform the principal component analysis (PCA) on the representations obtained from the pre-final layer output of the VGGNet. The reduced dimension representation is to be given as input to an MLFFNN with two hidden layers that use Tanh as the activation function. Use the softmax as the activation function in the output layer. Mini-batch mode with the number of subsets as 5 is used for fine-tuning. The AdaM method is the optimization method for fine-tuning.**

The report should include the Error vs Epoch plots, classification accuracies, and the confusion matrices for each of the models.